

Adaptive Methods in Clinical Research

Practical 5: Re-designing your own (adaptive) trial

Following major cardiac surgery, patients routinely receive supplemental oxygen following extubation. Despite major advances in cardiac surgery, patients frequently experience pulmonary complications and require intensive respiratory support following their procedure. Optimizing post-operative oxygen delivery is critical for improving patient comfort, minimizing complications, and accelerating recovery, thereby reducing hospital length of stay (LOS) and the number of hospital and intensive care unit re-admissions.

The intervention of interest is High-Flow Nasal Oxygen (HFNO), a less invasive method of respiratory support compared to traditional non-invasive ventilation (NIV). HFNO delivers heated and humidified oxygen at a high flow rate, potentially improving lung mechanics and patient compliance. Whilst on HFNO, patients are able to talk and eat freely.

Standard care is normally standard oxygen therapy via nasal prongs or a face mask. This oxygen is delivered low-flow, dry and not warmed.

A randomised open-label pilot study conducted at Papworth Hospital suggested that in high-risk patients who received HFNO post-cardiac surgery had a quicker recovery trajectory compared to patients who received standard oxygen therapy, demonstrating an average reduction of 2 days in their total hospital LOS compared to those receiving standard oxygen therapy. Unfortunately, it is not possible to blind patients or clinicians to the allocation of HFNO or standard oxygen therapy, which may lead to bias in the LOS endpoint.

We are asked to design a larger confirmatory study. The objective is to gather sufficient evidence of the potential of HFNO to improve the recovery post surgery in comparison to standard care, focusing on patient outcomes and resource use.

Some additional information from the Pilot is provided in the next page.

Working in groups of 4–5 people, propose your design for the stated problem. Specifically, answer the following:

- (a) How have you formally formulated the objective of the trial?
- (b) What is the primary endpoint that you would consider for this trial and why? List a few options and try to weigh in pros and cons.
- (c) For the stated problem, write down the design that you propose to use? Discuss whether it is adaptive or not, and what the adaptations are. What assumptions will you need to make to design such a study?

- (d) Propose scenarios/configurations and metrics to be used for the evaluation of the design?
- (e) If your design can be implemented with one of the designs learned/implemented in this course, find the design parameters and its operating characteristics. If the design is not readily implemented, could you use the practicals of this course to inform a ballpark of the sample size needed in such a trial?
- (g) Prepare a 5-minute presentation with the proposed design and the scheme to set up such a design.

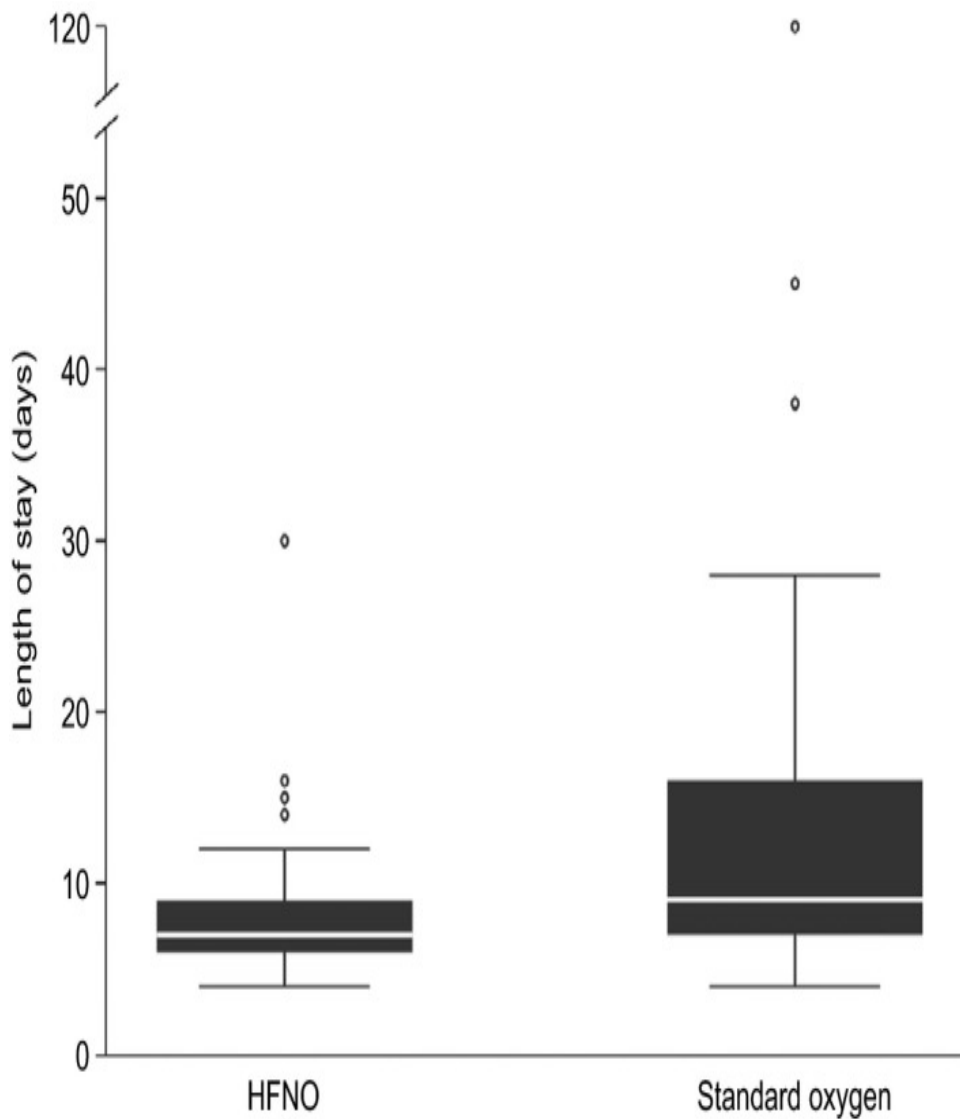


Table 2

Study end-points for patients randomly allocated to either the high-flow nasal oxygen group or standard oxygen group.

	High-flow nasal oxygen group	Standard oxygen group	p value
	n = 49	n = 45	
Total length of stay; days	7 (6–9 [4–30])	9 (7–16 [4–120])	0.012
ICU length of stay; days	1 (1–2 [1–15])	1 (1–2 [1–23])	0.949
<u>Re-admission to ICU</u>	<u>1 (2.0%)</u>	<u>7 (15.6%)</u>	<u>0.026</u>
Six-minute walk test; m	214 (116–280 [40–380])	165 (98–251 [60–510])	0.330
Lung function			
FVC; l	1.9 (1.6–2.3 [0.8–3.8])	1.9 (1.5–2.3 [0.9–3.5])	0.480
FVC; % of predicted value	57 (45–66 [31–102])	57 (47–69 [34–123])	0.990
FEV ₁ ; l	1.5 (1.1–1.7 [0.7–2.5])	1.2 (1.1–1.5 [0.5–2.7])	0.180
FEV ₁ ; % of predicted value	54 (42–64 [29–81])	53 (39–65 [18–83])	0.690

ICU, intensive care unit; FEV₁, forced expiratory volume in one second; FVC, forced vital capacity.

Notes:

If you have further questions, you can ask imposter/Twitter clinical experts present in the room. Just ask!